



Loud, Not Right

Validating a Smart Monitor’s Cry-Classification Label Against What Actually Soothed the Baby

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Background

Smart monitors now grade a cry’s likely cause and post a confidence score alongside the label. Caregivers act on both. Little is known about whether that confidence tracks the label’s correctness, or something else entirely.

Aims

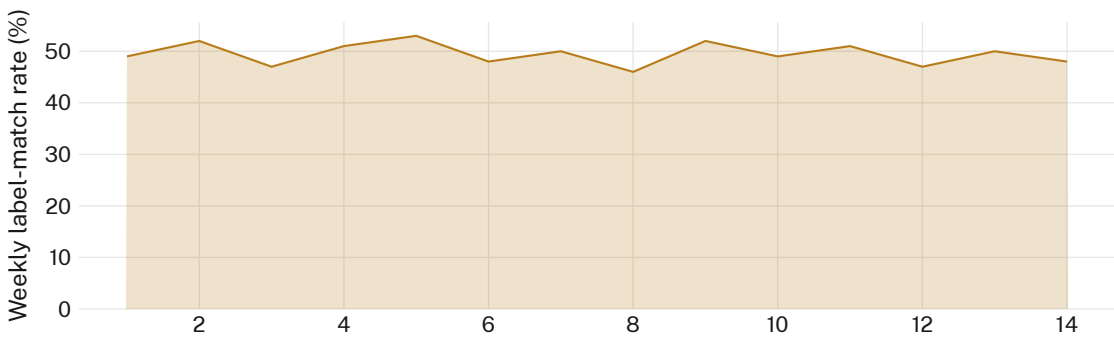
- Test whether the monitor’s predicted cry category matches the soothing action that actually settled the baby
- Test whether the monitor’s confidence score tracks label correctness or the cry’s acoustic loudness
- Track whether the match rate improves as caregivers log more cries over the trial

Methods

- 68 families · commercial monitor + companion logging app · cries logged blind to the on-device label · 14-week deployment
- 4-category taxonomy (hungry, tired, discomfort, overstimulated) matched against the caregiver’s own post-hoc read of what worked
- Confidence score regressed separately against label correctness and against cry loudness/duration; matching thresholds pre-registered

Results

Label-match rate held flat across the deployment, while confidence tracked the cry’s loudness far more closely than whether the label was right.



Label-match rate held between 47% and 53% across all 14 weeks of the trial (n = 68 families), with no upward drift as caregivers logged more cries.

Discussion & conclusions

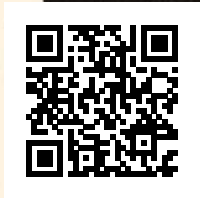
A confident label is not the same claim as a correct one, and the interface does not currently distinguish them. Caregivers who trust the score are trusting the cry’s volume more than its content. Next: testing whether hiding the confidence display changes which soothing action a caregiver reaches for first.

References

1. Micheletti, M., Yao, X., Johnson, M., & de Barbaro, K. (2022). Validating a model to detect infant crying from naturalistic audio. *Behavior Research Methods*, 55(6), 3187–3197. doi:10.3758/s13428-022-01961-x
2. Hong, M., Zhang, C. J., Yang, L., Song, Y., & Jiang, D. (2024). InfantCryNet: A data-driven framework for intelligent analysis of infant cries. *arXiv:2409.19689*
3. Ji, C., Chen, M., Li, B., & Pan, Y. (2021). Infant cry classification with graph convolutional networks. *arXiv:2102.02909*
4. Liu, Q., Song, L., Xu, D., & Long, Y. (2024). ICSD: An open-source dataset for infant cry and snoring detection. *arXiv:2408.10561*
5. Tan, W., Yao, Q., & Liu, J. (2023). Weakly supervised detection of baby cry. *arXiv:2304.10001*
6. Chikere, I., & Umbile, P. (2026). Colour-Coded, Not Calmer: A Survey of Reset-Chart Completion and Household Dispute Rates in the First Week of Term. School of Emergent Priorities, Slop University. doi:10.5555/slop.azf7r2

Ethics approval 2025/162; audio processed on-device only, no recordings retained beyond the labelling window.

We thank the families of the monitored nurseries. Office of Research Outputs · slop.university



“Confidence climbed. Correctness didn’t.”

Label-match rate held between 47% and 53% across all 14 weeks of the trial (n = 68 families).